

Unit 4, Lesson 9: Discovering Relationships of Parallel and Perpendicular Lines



Benchmark

MA.912.AR.2.3 Write a linear two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.



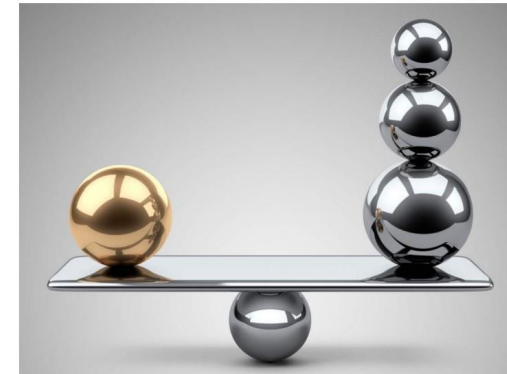
Learning Targets

- ☐ I can discover the relationship between parallel lines.
- ☐ I can discover the relationship between perpendicular lines.



4.9.1 Warm-Up: Notice and Wonder

1. What do you notice?



2. What do you wonder?

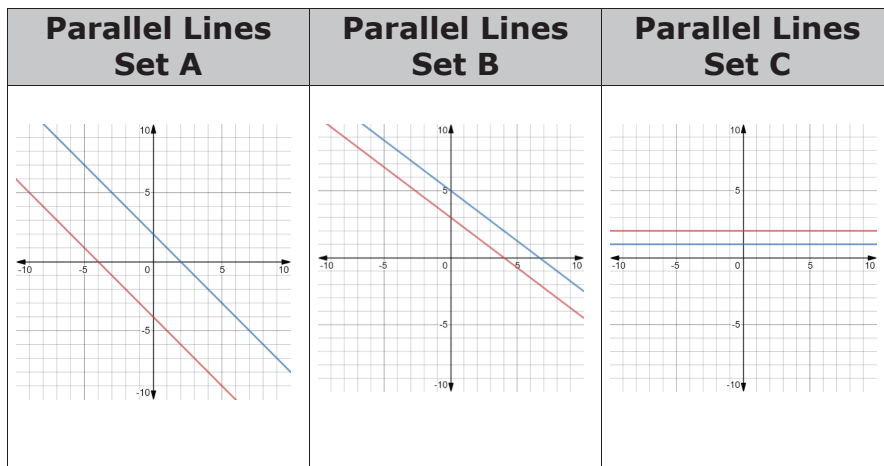
3. Write a question about the picture for the class.



4.9.2 Exploration: Parallel Lines

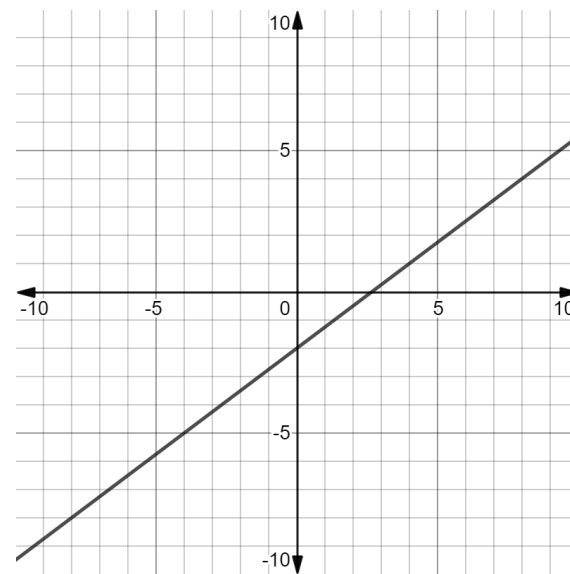
- Examine the pairs of parallel lines.

Write down your observations and questions.



Observations	Questions

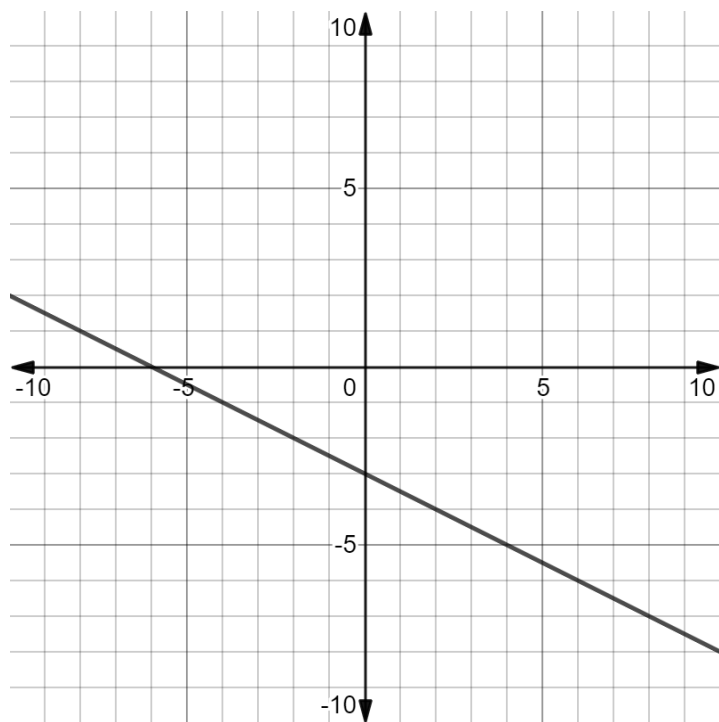
- The graph of $y = \frac{3}{4}x - 2$ is shown on the coordinate grid.



- Draw a line that is parallel to $y = \frac{3}{4}x - 2$.
- Use the slope and the y -intercept of the line you drew to write the equation of a line that is parallel to $y = \frac{3}{4}x - 2$.
- Write the equations of the lines that two other classmates wrote.

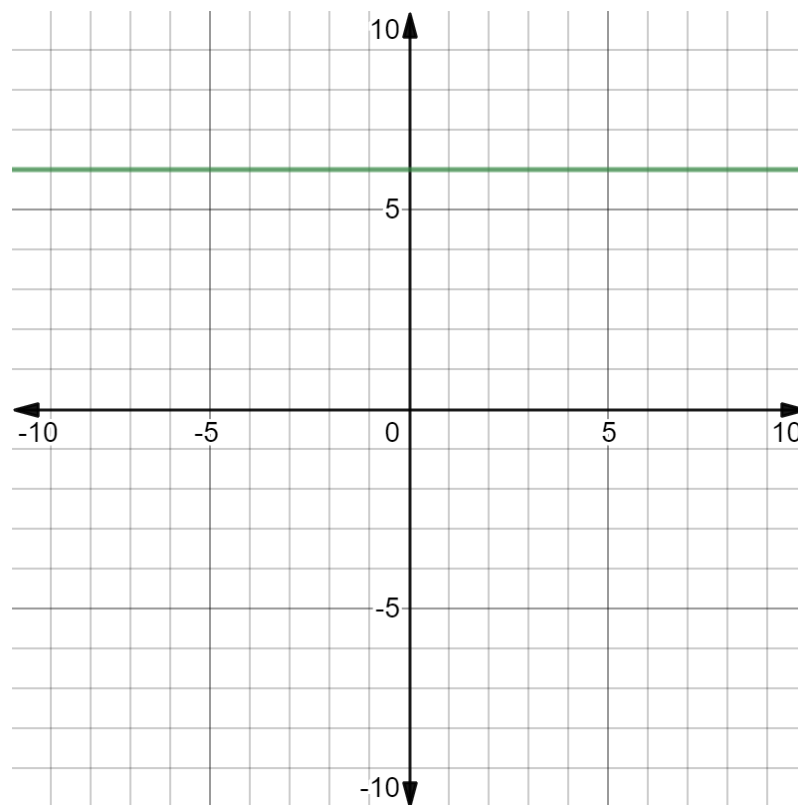
Classmate	Equation

3. The graph of $y = -\frac{1}{2}x - 3$ is shown on the coordinate grid.



- Sketch a line that is parallel to $y = -\frac{1}{2}x - 3$ and passes through the point $(3, 4)$.
- Write the equation of the line in point-slope form.

4. The graph of $y = 6$ is shown on the coordinate grid.

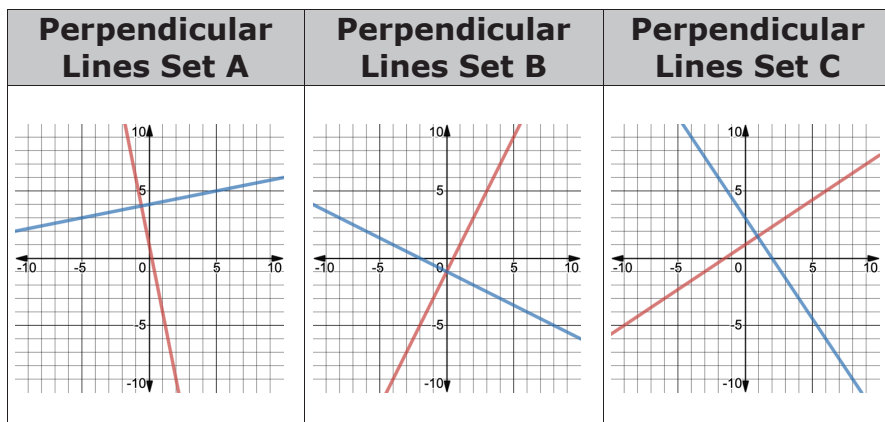


- Sketch a line that is parallel to $y = 6$ and passes through the point $(-2, -4)$.
 - Write the equation of the line.
5. Write a conjecture about the slopes of parallel lines.



4.9.3 Exploration: Perpendicular Lines

- Examine the pairs of perpendicular lines.



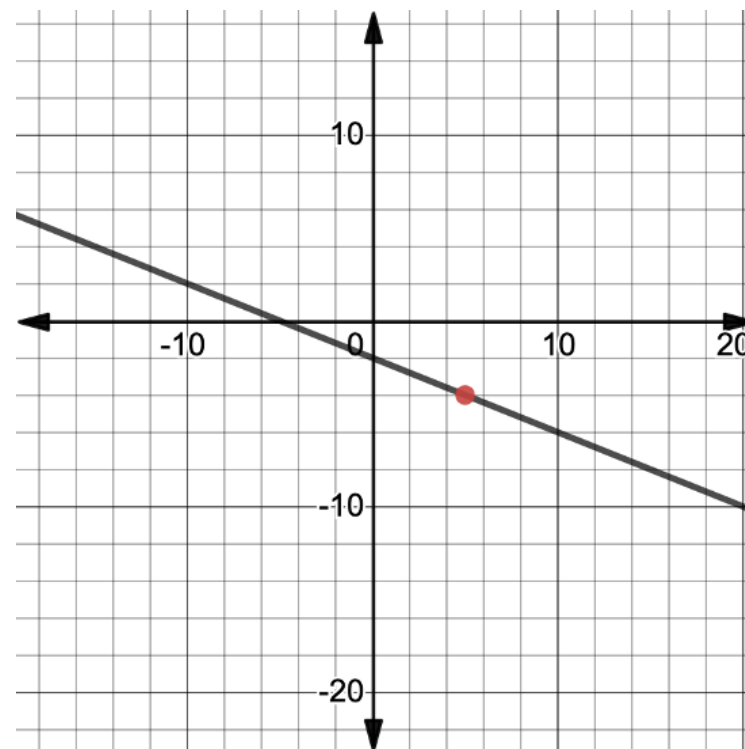
- Write down your observations and questions.

Observations	Questions

- Record your partner's observations and questions.

Observations	Questions

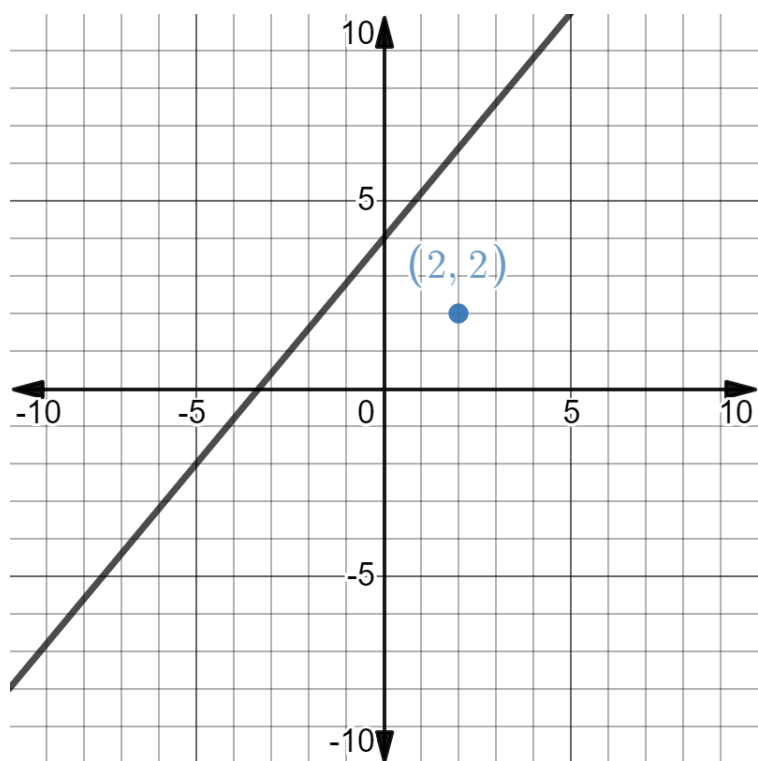
- The graph of $y = -\frac{2}{5}x - 2$ is shown on the coordinate grid.



- Draw a line that is perpendicular to $y = -\frac{2}{5}x - 2$ and goes through the point $(5, -4)$ by using the corner of a sheet of paper for a right angle.
- Use your observations and the line you drew to determine the slope that completes the equation of the line that is perpendicular to $y = -\frac{2}{5}x - 2$ and passes through the point $(5, -4)$.

$$y + 4 = \underline{\hspace{2cm}}(x - 5)$$

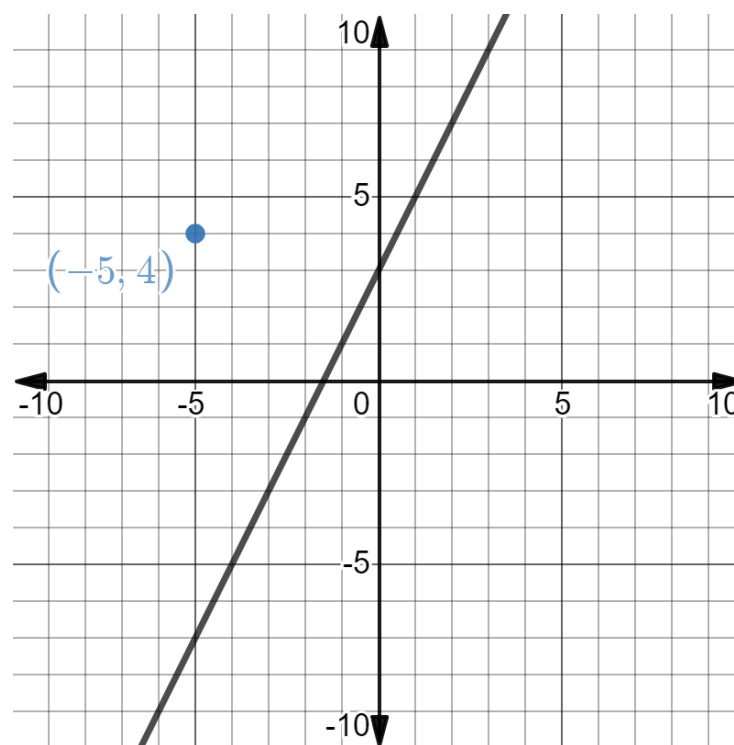
3. The graph of $6x - 5y = -20$ is shown on the coordinate grid.



- Draw a line that is perpendicular to $6x - 5y = -20$ and goes through the point $(2, 2)$ by using the corner of a sheet of paper for a right angle.
- Use your observations and the line you drew to determine the slope that completes the equation of the line that is perpendicular to $6x - 5y = -20$ and passes through the point $(2, 2)$.

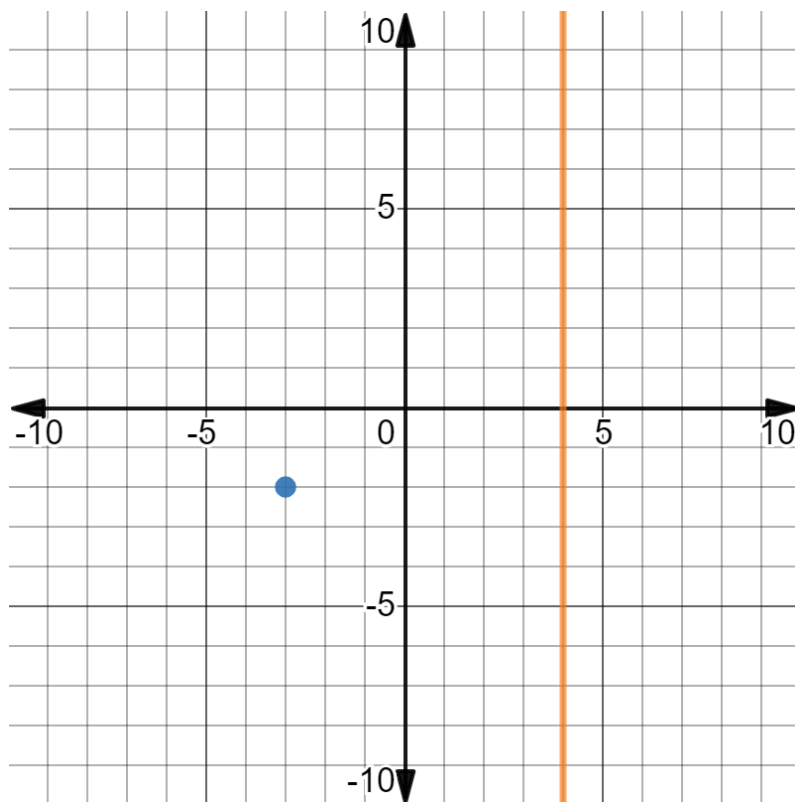
$$y - 2 = \underline{\hspace{1cm}}(x - 2)$$

4. The graph of $y - 5 = 2(x - 1)$ is shown on the coordinate grid.



- Draw a line that is perpendicular to $y - 5 = 2(x - 1)$ and goes through the point $(-5, 4)$ by using the corner of a sheet of paper for a right angle.
- Write the equation of the line that is perpendicular to $y - 5 = 2(x - 1)$ at the point $(-5, 4)$.

5. The graph of $x = 4$ is shown on the coordinate grid.



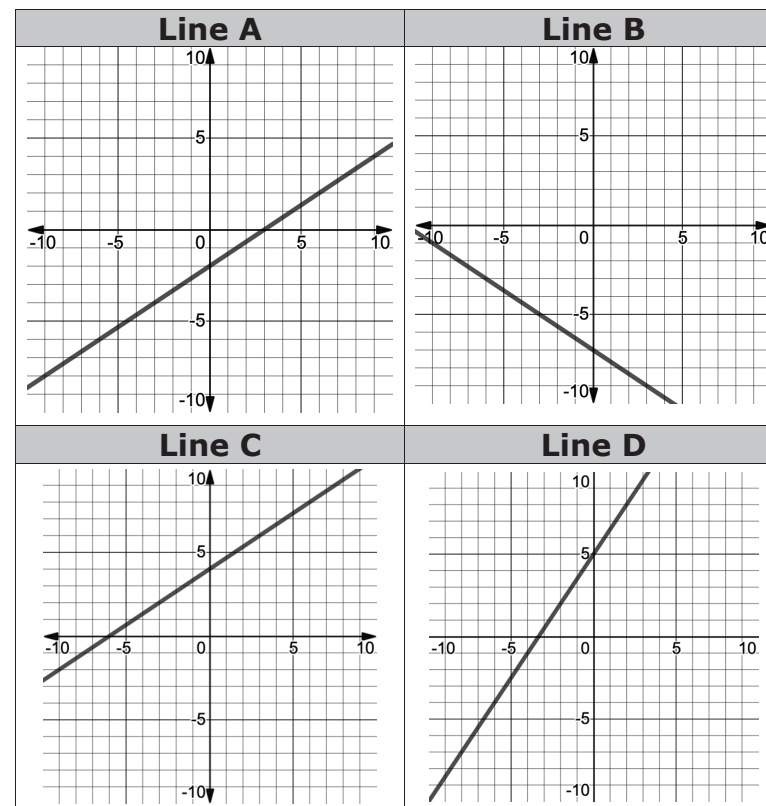
- Draw a line that is perpendicular to $x = 4$ and goes through the point $(-3, -2)$.
 - Write the equation of the line that is perpendicular to $x = 4$ that passes through the point $(-3, -2)$.
6. Write a conjecture about the slopes of perpendicular lines.



4.9.4 Wrap-Up: Ticket Out the Door

1. Match the two lines that are parallel and the two lines that are perpendicular. Then, match the equations.

	Lines	Equations
Parallel		
Perpendicular		



Equation E	Equation F
$y + 2 = \frac{2}{3}(x + 9)$	$2x + 3y = -21$
Equation G	Equation H
$-2x + 3y = -6$	$y = \frac{3}{2}x + 5$